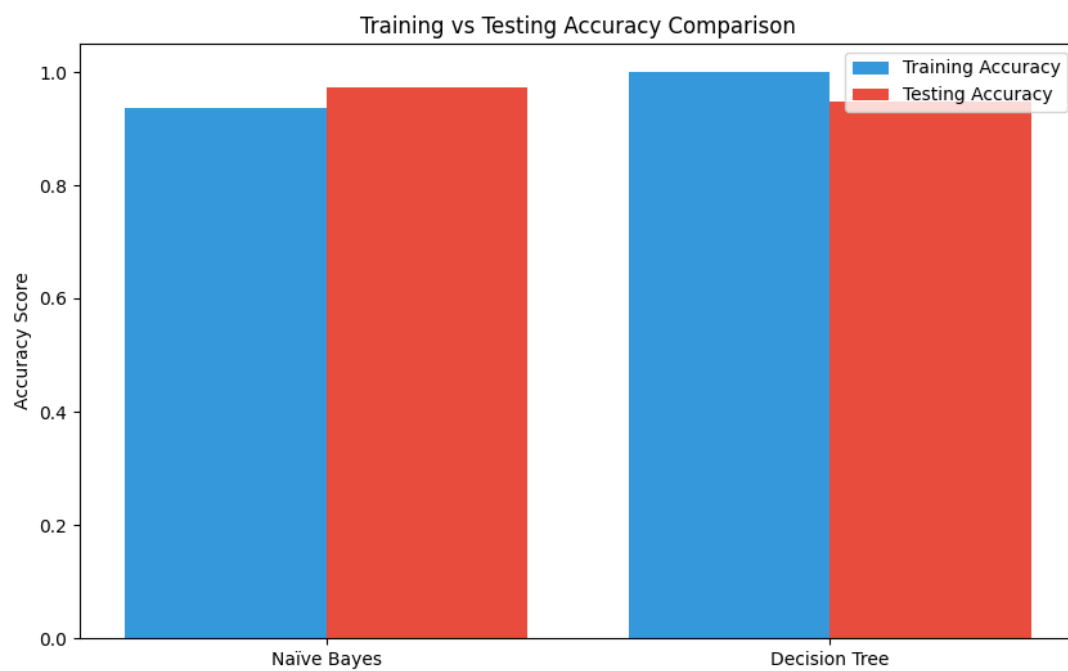
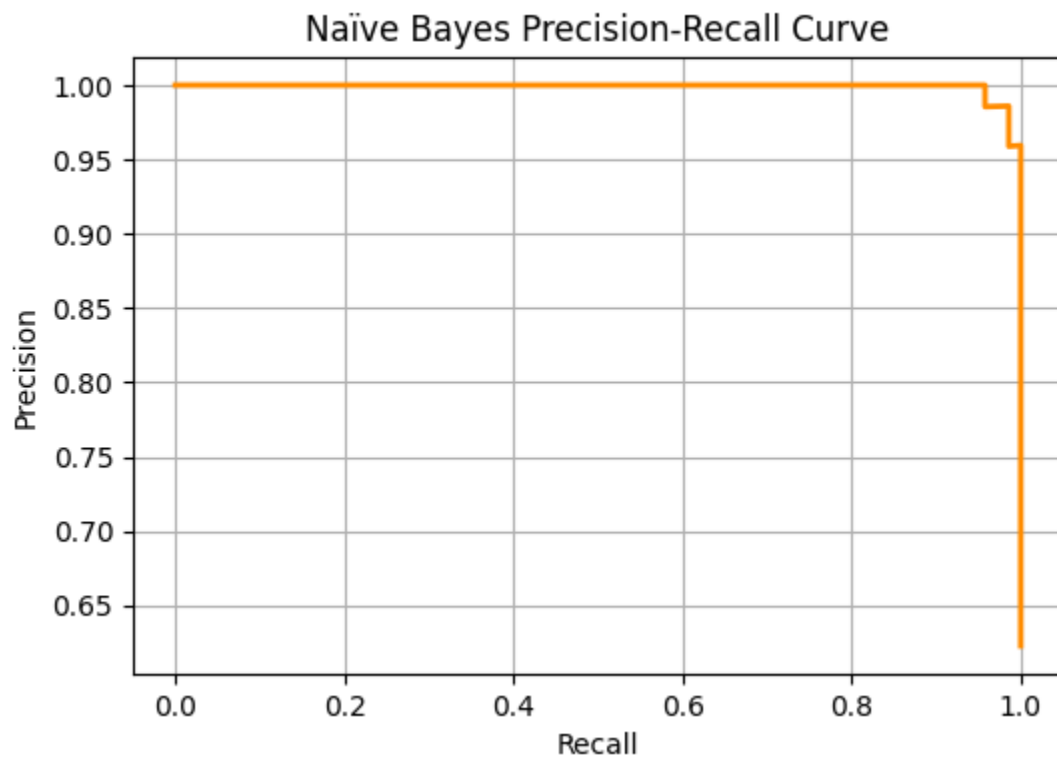
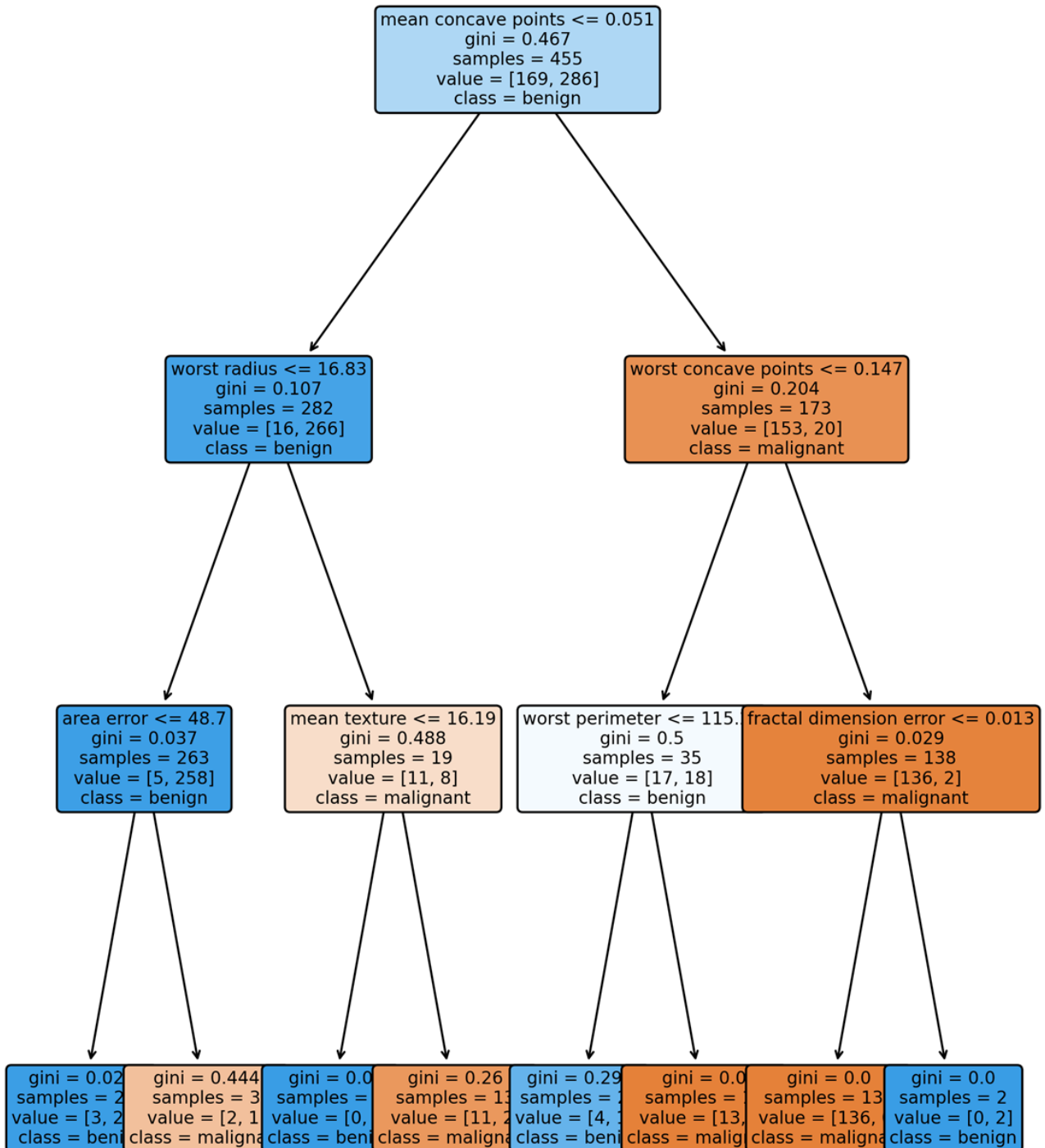




Decision Tree (Max Depth = 3)



```
• > python l5_e1.py; python3 l5_e2.py; python l5_e3.py
```

```
--- Experiment 1: Naïve Bayes Results ---
```

```
Accuracy: 0.9737
```

```
Confusion Matrix:
```

```
[[40  3]
 [ 0 71]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	1.00	0.93	0.96	43
1	0.96	1.00	0.98	71
accuracy			0.97	114
macro avg	0.98	0.97	0.97	114
weighted avg	0.97	0.97	0.97	114

```
/home/abhinav/kaam/MLLab/l5_e1.py:34: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
```

```
plt.show()
```

```
--- Experiment 2: Decision Tree Results ---
```

```
Accuracy: 0.9474
```

```
Confusion Matrix:
```

```
[[40  3]
 [ 3 68]]
```

```
Tree visualization saved as ex2_decision_tree.png
```

```
PR Curve saved as ex2_pr_curve.png
```

```
Saved: ex3_accuracy_comparison.png
```

```
Saved: ex3_roc_comparison.png
```

```
Saved: ex3_cm_heatmaps.png
```

